Mini Project

A/B TEST ANALYSIS FOR AN E-COMMERCE WEBSITE
Comparing Two Pricing Strategies Using Hypothesis Testing

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Introduction

E-commerce websites regularly try different designs, pricing pages and offers to improve their sales. However, before applying a new change to all customers, it is important to check whether the change actually makes a difference.

For this purpose, **A/B testing** is commonly used.

In an A/B test, users are divided into two groups and each group is shown a different version of a website.

For this project, the two groups are:

- **Control Group:** Users who see the existing or old pricing page.
- **Treatment Group:** Users who see the new pricing page.

The dataset also records whether a user completed a purchase. A value of **1** means that the user converted, while **0** means that the user did not convert

Dataset Description

A/B Test Dataset

The file ab_data.csv contains **294,478 records**. It provides information about the users who participated in the experiment.

The main columns are:

Column	Description
`user_id`	Unique ID of the user
`timestamp`	Time at which the user participated
`group`	Control or Treatment
`landing_page`	Old page or New page
`converted`	Purchase made or not

The converted column is the main variable used to measure the success of the two strategies.

- 1 → User made a purchase
- 0 → User did not make a purchase

Country Dataset

The countries.csv file contains the country information of the users. The dataset includes users from:

- United States (US)
- United Kingdom (UK)
- Canada (CA)

The two datasets are connected using the user_id column.

Data Cleaning

Before starting the statistical analysis, the dataset was checked for errors and duplicate records. This step is important because incorrect or repeated records can affect the final results.

Group and Landing Page Mismatch

Ideally, the data should follow this pattern:

Control → Old Page

Treatment → New Page

However, some records did not follow this relationship. For example, some users belonging to the Control group were recorded as seeing the new page.

There were **3,893 such mismatched records**, and these records were removed from the analysis.

Duplicate Users

The dataset also contained users who appeared more than once. If the same user is counted multiple times, it can unnecessarily affect the conversion rate.

There were **3,894 duplicate user IDs**. Only the first recorded visit of each user was retained.

After cleaning the data and merging it with the country dataset, the final dataset contained **290,584 users**.

Group	Number of Users	Conversions
Control	145,274	17,489
Treatment	145,310	17,264
Total	**290,584**	**34,753**

Tests

INDEPENDENT T-TEST

The first test compares the average conversion of the Control and Treatment groups.

Since the converted column contains only 0 and 1, its average directly represents the conversion rate of a group.

The project uses the **Welch independent two-sample t-test**.

Null Hypothesis

$$H_0: \mu_{Treatment} = \mu_{Control}$$

This means that there is no significant difference between the conversion rates.

Alternative Hypothesis

$$H_1: \mu_{Treatment} \neq \mu_{Control}$$

This means that there is a significant difference between the two conversion rates.

The t-test gives us a **t-statistic and p-value**, which are used to make the final decision.

CHI-SQUARE TEST

The second test is the **Chi-square test of independence**.

This test is useful because both variables being compared are categorical:

- Pricing strategy → Control / Treatment
- Conversion → Converted / Not Converted

The test checks whether the conversion result is related to the pricing strategy.

Null Hypothesis

$$H_0:$$

Pricing strategy and conversion are independent.

Alternative Hypothesis

$$H_1:$$

Pricing strategy and conversion are related.

The following contingency table is used:

Group	Not Converted	Converted
Control	127,785	17,489
Treatment	128,046	17,264

ONE-WAY ANOVA

The third technique used is **One-way ANOVA**.

The purpose of ANOVA in this project is to check whether the conversion rate changes significantly across different countries.

The countries considered are:

- Canada
- United Kingdom
- United States

ANOVA is also performed on the six combinations of country and pricing strategy.

For example:

- Canada – Control
- Canada – Treatment
- UK – Control
- UK – Treatment
- US – Control
- US – Treatment

Null Hypothesis

$$H_0: \mu_1 = \mu_2 = \ldots = \mu_k$$

This means that the average conversion rate is the same across the groups.

Alternative Hypothesis

$$H_1:$$

At least one group has a different average conversion rate.

Confidence Interval

Apart from calculating p-values, a **95% confidence interval** is also used.

A confidence interval gives us a reasonable range within which the actual difference between the two strategies may lie.

The calculated confidence interval for the difference between Treatment and Control is approximately:

$$[-, 0.394\%0.078\%]$$

Since this range includes **0**, the data does not give strong evidence of a real difference between the two strategies.

Results

DESCRIPTIVE OVERVIEW

The conversion rate is calculated using:

$$Conversion\ Rate = \frac{Number\ of\ Conversions}{Total\ Users} \times 100$$

The results obtained after cleaning are:

Pricing Strategy	Users	Conversions	Conversion Rate
Control / Old Page	145,274	17,489	12.039%
Treatment / New Page	145,310	17,264	11.881%

The difference between the two conversion rates is:

$$11.881 - 12.039 = -0.158$$

So, the Treatment group has an observed conversion rate approximately **0.158 percentage points lower** than the Control group.

However, this difference by itself does not tell us whether the new strategy is actually worse. We need hypothesis testing to determine whether the difference is statistically meaningful.

Figure 1: Users in Control and Treatment Groups

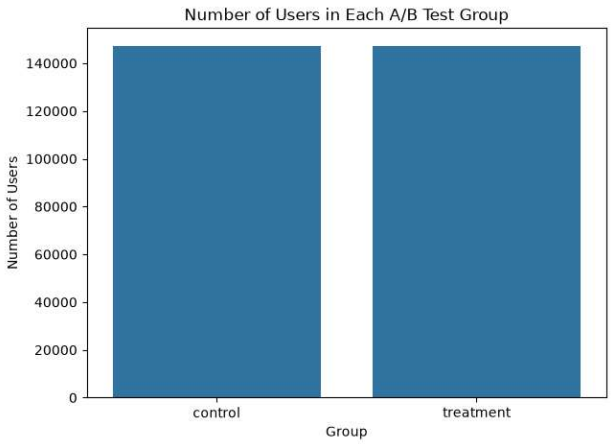
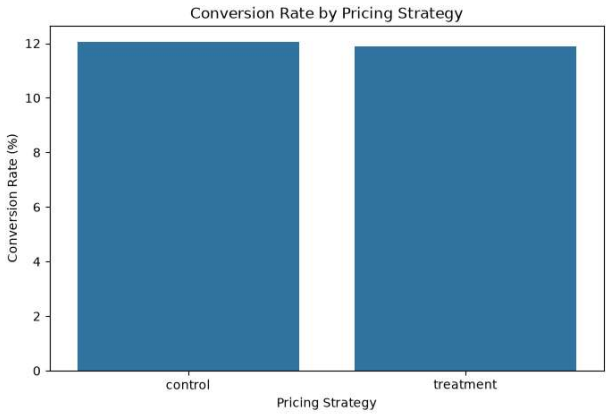


Figure 2: Conversion Rate by Pricing Strategy



T-TEST RESULTS

The Welch t-test gave the following results:

Measure	Result
Control conversion	12.039%
Treatment conversion	11.881%
Difference	−0.158 percentage points
t-statistic	−1.311
p-value	0.190

The p-value is:

$$0.190 > 0.05$$

Therefore, we **fail to reject the null hypothesis**.

In simple words, the difference seen between the two conversion rates is not statistically significant according to the t-test.

The 95% confidence interval is approximately:

$$[-, 0.394\%0.078\%]$$

Since zero falls within this interval, the result also supports the conclusion that a significant difference has not been established.

CHI-SQUARE TEST RESULTS

The Chi-square test was performed using the group and conversion results.

The results are:

Measure	Result
Chi-square statistic	1.704
Degrees of freedom	1
p-value	0.192

Here:

$$0.192 > 0.05$$

Therefore, the null hypothesis is not rejected.

This means that, based on this dataset, there is no statistically significant evidence that conversion depends on whether the user saw the old or new pricing page.

ANOVA RESULTS

The country-wise conversion rates are shown below:

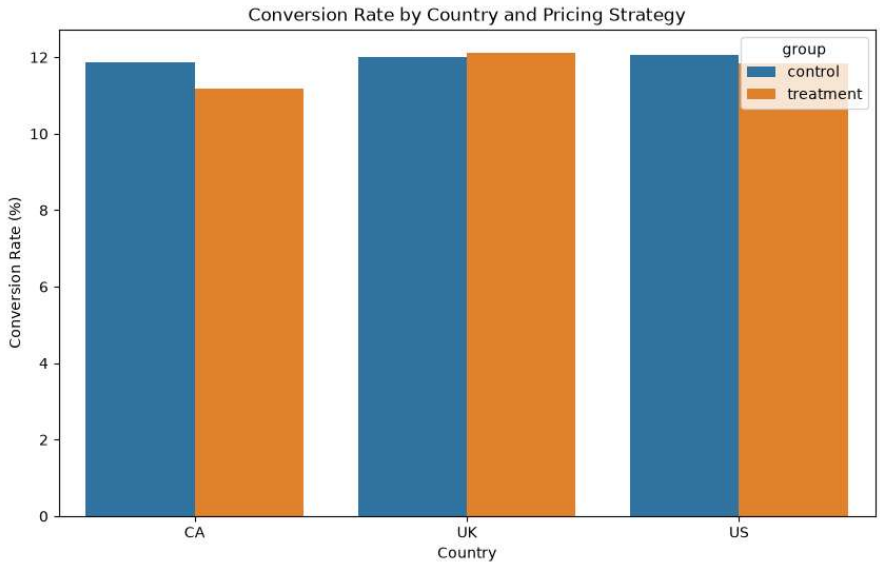
Country	Users	Conversion Rate
Canada	14,499	11.53%
United Kingdom	72,466	12.06%
United States	203,619	11.95%

The ANOVA results are:

Comparison	F-statistic	p-value
Three countries	1.605	0.201
Six country-strategy groups	1.466	0.197

Both p-values are greater than 0.05. Therefore, there is no statistically significant evidence that conversion rates are different across the countries or across the country-strategy combinations.

Figure 3: Conversion Rate by Country and Pricing Strategy



Discussion & Conclusion

DISCUSSION

After performing all the statistical tests, the results are quite consistent.

The t-test produced a p-value of **0.190**, while the Chi-square test produced a p-value of **0.192**. Both values are greater than the selected significance level of 0.05.

The ANOVA results also gave p-values above 0.05 for both the country comparison and the country-strategy comparison.

The actual conversion rates were slightly different:

- **Control:** 12.04%
- **Treatment:** 11.88%

At first glance, the Control group appears to have a slightly higher conversion rate. However, the statistical tests show that the observed difference is not large enough to conclude that the two strategies genuinely perform differently based on this experiment.

It is also important to understand that a result being **"not statistically significant" does not mean that the two pricing pages are exactly the same**. It simply means that this experiment does not provide enough evidence to establish a statistically significant difference.

The large sample size makes the analysis useful for detecting moderate differences, although very small effects may still go undetected.

CONCLUSION

This project used A/B testing and statistical hypothesis testing to compare an existing pricing page with a new pricing page on an e-commerce website.

After cleaning the data, **290,584 users** were included in the final analysis.

The Control group had a conversion rate of **12.04%**, whereas the Treatment group had a conversion rate of **11.88%**.

The statistical results were:

Test	p-value	Result
Welch T-Test	0.190	Not significant
Chi-square Test	0.192	Not significant
ANOVA – Country	0.201	Not significant
ANOVA – Country & Strategy	0.197	Not significant

All the p-values are greater than 0.05. Therefore, the analysis does not provide sufficient statistical evidence that the new pricing page improves conversion compared with the existing page